

Python Interview Questions

Day 03 • Questions, Answers & Examples

1. What is the difference between a list and a tuple in Python?

Answer: A list is mutable, meaning its elements can be changed after creation. A tuple is immutable, meaning its elements cannot be changed after creation.

Example:

```
my_list = [10, 20, 30]
my_list[0] = 100
print(my_list)
```

```
my_tuple = (10, 20, 30)
print(my_tuple)
```

Output:

```
[100, 20, 30]
(10, 20, 30)
```

2. What is a string in Python?

Answer: A string is a sequence of characters enclosed in single quotes, double quotes, or triple quotes. Strings are immutable in Python.

Example:

```
name = "Python"
print(name)
print(name[0])
```

Output:

```
Python
P
```

3. What is indexing in Python?

Answer: Indexing is the process of accessing individual elements of a sequence using their position. Python uses zero-based indexing, so the first element has index 0.

Example:

```
language = "Python"
print(language[0])
print(language[3])
```

Output:

```
P
h
```

4. What is slicing in Python?

Answer: Slicing is used to extract a portion of a sequence such as a string, list, or tuple. The general syntax is `sequence[start:stop:step]`.

Example:

```
text = "Python"
print(text[0:3])
print(text[2:5])
```

Output:

```
Pyt  
tho
```

5. What is the difference between mutable and immutable objects in Python?

Answer: Mutable objects can be changed after they are created, while immutable objects cannot be changed after creation. Lists, sets, and dictionaries are mutable; strings, integers, and tuples are immutable.

Example:

```
numbers = [1, 2, 3]  
numbers.append(4)  
print(numbers)  
  
name = "Python"  
# name[0] = "J" # causes an error
```

Output:

```
[1, 2, 3, 4]
```

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